

Here are some fields: $\mathbb{F}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}$, $\mathbb{Z}/5\mathbb{Z}$, $\mathbb{Z}/p\mathbb{Z}$.

For a custom letter in “blackboard bold”: $\mathbb{P}$

Here are some vector spaces: V , W , $\mathbb{R}^2$, $\mathbb{C}^n$, $M_{3 \times 3}^{\mathbb{R}}$

Some symbols for vectors, in bold: $\mathbf{v}$, $\mathbf{w}$, $\mathbf{0}$, or for a vector with your choice of letter: $\mathbf{q}$

Here are some important symbols: $\cdot$ $\times$ $\rightarrow$ $\{$ $\}$ $\cup$ $\cap$ span dim det π $\lambda \in \subset \sum_{i=1}^n$

You can make a nice displayed equation; some things (like sums) are less cramped this way:

$$\sum_{i=1}^k a_i \mathbf{v}_i = 0$$

Here is a vector: $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

And a linear combination: $\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$

Now with subscripts!

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \cdots + a_k \mathbf{v}_k = 0$$

For subscripts and superscripts that are longer than one character, you need to surround them with braces: $a_{j,k}$. But if they’re just one character long, the braces aren’t needed (but they don’t hurt, either).

Here’s a half inch vertical skip:

Here is a matrix: $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Here are an inverse and a transpose:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

How about nicely aligned system of equations?

$$\begin{array}{rcl} x + 2y + 3z & = & 0 \\ x + 3y + 4z & = & 0 \end{array}$$